Advanced information, computation, communication I
EPFL - Fall semester 2026-2027

Week 4

October 2, 2026

Exercise 1. Let $A = \{x \mid x > 0\}$, $B = \{x \mid x \neq 1\}$, and $C = \{x \mid x < 1\}$.

Is the following equality correct?

$$(A \cap B) \cup C = A \cup (B \cap C)$$

Exercise 2. Let $E = A \cap (B - C)$ and $F = (A \cap B) - (A \cap C)$ Which of the following propositions are always true?

- ◯ $E \subseteq F$
- ◯ $F \subseteq E$
- ◯ $E = F$
- ◯ None of the other answers are always true.

Hint: $A - B = A \cap \bar{B}$

Exercise 3. Let $f : \{x \mid x \in \mathbf{R}, -2 \leq x \leq 5\} \to \mathbf{R}$,

$$x \mapsto \begin{cases} 3 + \frac{3}{2}x & \text{for} \quad -2 \leq x \leq 0 \\ \lfloor x \rfloor & \text{for} \quad 0 \leq x < 2 \\ x^2 & \text{for} \quad 2 \leq x \leq 5. \end{cases}$$

- ◯ f is injective but not surjective.
- ◯ f is surjective but not injective.
- ◯ f is bijective.
- ◯ f is not a function.

Exercise 4. Let $f : \{x \mid x \in \mathbf{R}, 0 < x < 1\} \to \mathbf{R}$,

$$x \mapsto \begin{cases} 2 - \dfrac{1}{x} & \text{if} \quad 0 < x < 1/2 \\ \dfrac{1}{1-x} - 2 & \text{if} \quad 1/2 \leq x < 1. \end{cases}$$

- ◯ f is not injective and not surjective.
- ◯ f is injective but not surjective.
- ◯ f is surjective but not injective.
- ◯ f is bijective.

Exercise 5. Determine which of these functions are bijections from $\mathbf{R}$ to $\mathbf{R}$

◯ $f(x) = 2x + 1$

◯ $f(x) = x^2 + 1$

◯ $f(x) = x^3$

◯ $f(x) = (x^2 + 1)/(x^2 + 2)$

Exercise 6. For any $\delta \in \mathbf{R}$ let f_δ and g_δ be the following two functions from $\mathbf{R}$ to $\mathbf{R}$

$$f_\delta(x) = \begin{cases} x + \delta & \text{if } x \in \mathbf{Z} \\ -x + \delta & \text{if } x \notin \mathbf{Z}, \end{cases} \qquad g_\delta(x) = \begin{cases} x + \delta & \text{if } x \in \mathbf{Z} \\ -x - \delta & \text{if } x \notin \mathbf{Z}. \end{cases}$$

Consider the two statements

$$\forall \delta \in \mathbf{R} \;\; f_\delta \text{ is a bijection} \qquad \text{and} \qquad \forall \delta \in \mathbf{R} \;\; g_\delta \text{ is a bijection.}$$

◯ Only the second statement is true.

◯ Only the first statement is true.

◯ They are both true.

◯ They are both false.

Exercise 7. Let $\mathcal{P}(X)$ denote the power set of a set X and let $\emptyset$ denote the empty set. Consider the two statements

for any sets A and B, if $\mathcal{P}(A) = \mathcal{P}(B)$, then $A = B$;

and

there exists a set C such that $\mathcal{P}(C) = \emptyset$.

◯ They are both true.

◯ Only the first is true.

◯ Only the second is true.

◯ They are both false.

Exercise 8. Let $X = \{1, 2, 3, 4, 5\}$ and let $\mathcal{P}(X)$ denote the power set of X. Given the statements

$$\emptyset \in \mathcal{P}(X) \qquad\qquad \{\emptyset\} \in \mathcal{P}(X)$$

◯ Only the first is true.

◯ They are both true.

◯ Only the second is true.

◯ They are both false.